DATA SCIENCE LABORATORY MANUAL

Module V ? Advanced Data Science

Topic: Ensemble Learning

OVERVIEW & LEARNING OBJECTIVES

Bagging with Random Forest, Boosting algorithms (AdaBoost, Gradient Boost, XGBoost), and voting classifiers.

KEY TOPIC CONCEPTS & METHODOLOGY

- Understanding Ensemble Learning in modern Data Science workflows.
- Detailed breakdown: Bagging with Random Forest, Boosting algorithms (AdaBoost, Gradient Boost, XGBoost), and voting cla
- Mathematical formulas, code examples, and laboratory execution guidelines.
- Common pitfalls, edge cases, and best practices in practical applications.

LABORATORY PRACTICE / IMPLEMENTATION NOTES

1. Follow standard Python and Data Science coding practices.
2. Document all exploratory findings and statistical hypotheses.
3. Keep datasets and outputs organized in your repository workspace.